Lesson 13.3 - How to Calculate the Covariance

> **Video:** How To Calculate The Covariance Between X and Y by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

Formula

$$\operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1} \quad \text{(sample)}$$

For a **population**, divide by N instead of $n-1$.

Steps (table method)

1. Columns: x, y, $x-\bar{x}$, $y-\bar{y}$, $(x-\bar{x})(y-\bar{y})$.
2. Find $\bar{x}$ and $\bar{y}$.
3. Fill in the deviation columns. Each one sums to 0, which is a useful check.
4. Multiply the deviations row by row and add the products.
5. Divide by $n-1$.

Example 1 - positive covariance

x	y	$x-\bar{x}$	$y-\bar{y}$	product
2	3	−4	−7.2	28.8
4	7	−2	−3.2	6.4
6	10	0	−0.2	0
8	14	2	3.8	7.6
10	17	4	6.8	27.2
Sum 30	**Sum 51**	0	0	**70**

$\bar{x} = \sum x/n = 30/5 = 6,\ \ \bar{y} = \sum y/n = 51/5 = 10.2$

$$\operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1} = \frac{70}{5-1} = 17.5$$

Example 2 - negative covariance

x	y	$x - \bar{x}$	$y - \bar{y}$	product
3	20	−6	7.4	−44.4
6	17	−3	4.4	−13.2
9	13	0	0.4	0
12	9	3	−3.6	−10.8
15	4	6	−8.6	−51.6
Sum 45	**Sum 63**	0	0	**−120**

$\bar{x} = \sum x/n = 45/5 = 9,\ \bar{y} = \sum y/n = 63/5 = 12.6$

$$\text{cov}(x, y) = \frac{\sum(x - \bar{x})(y - \bar{y})}{n - 1} = \frac{-120}{5 - 1} = -30$$

(The video has a few slips while adding up this column, such as "−1.8" and "−20", but its final answer, −30, is correct.)

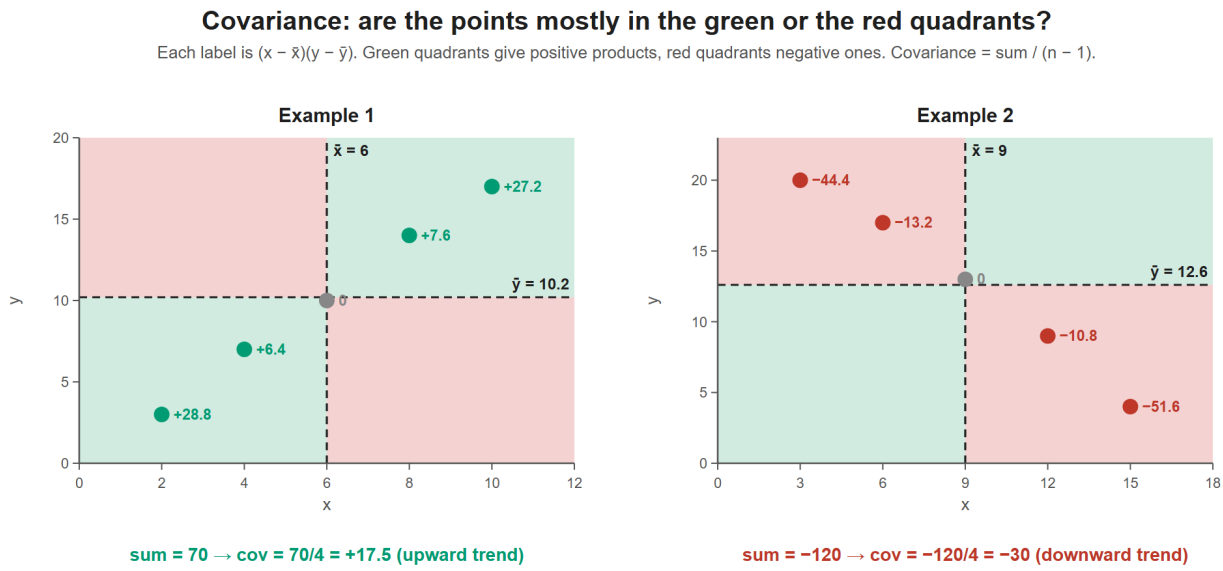
What the sign means

On a scatter plot:

- Example 1: as x increases, y increases (an upward trend) → covariance **positive** (+17.5).
- Example 2: as x increases, y decreases (a downward trend) → covariance **negative** (−30).

Covariance	Relationship
> 0	Positive: the variables move **together**
< 0	Negative: when one goes up, the other goes **down**
≈ 0	No linear relationship

> The **size** of the covariance depends on the units of x and y, so it is hard to say whether 17.5 is "big". The **correlation coefficient** (Lesson 13.4) rescales it to the range −1 to +1.



Points up-right or down-left of the means (green) add positive products. Points up-left or down-right (red) add negative products. Example 1 is all green, so the covariance is positive. Example 2 is all red, so it is negative.

Key takeaways

- Sample covariance $= \sum(x - \bar{x})(y - \bar{y})/(n - 1)$.
- The **sign** gives the direction of the relationship. The size is unit-dependent.
- The deviation columns always add to 0, which is a good check on your arithmetic.